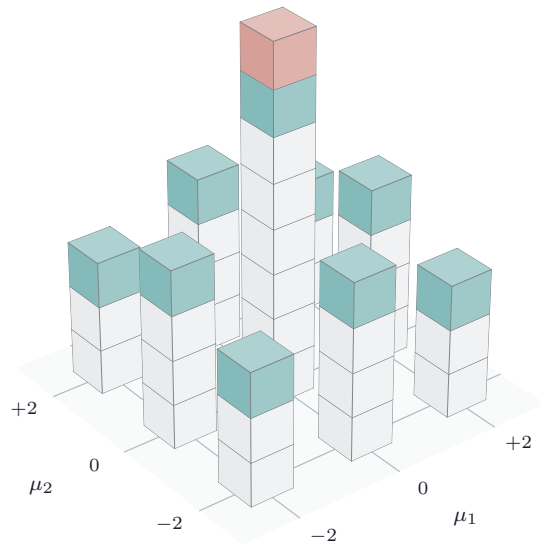




tangent space at Y of the Hodge locus of ω_a

■ ω_a exceptional: $\mathbb{C}\kappa$, dim 1

■ $\alpha \in \mathbb{Q}$: contains $T_Y\{X' \times X'\}$, dim 10

■ $T_Y\mathfrak{S} = \mathfrak{sp}^{-1,1}$: dim 36

blocks: weights (μ_1, μ_2) of a maximal torus of $\mathrm{SL}(V_1) \times \mathrm{SL}(V_2)$

	$E_3^-, \dim 1$	$E_{13}^-, \dim 3$	$E_{23}^-, \dim 3$	$E_{123}^-, \dim 9$
$H^1(T_Y)$ diagonal	$b_{11} = b_{22}: \kappa$	$a_1^2 - a_3^2$	$a_1^2 - a_2^2$	a_1, a_2, a_3 distinct
$H^1(T_Y)$ mixed	$\frac{3}{2}(a_2 - a_3)$	$-\frac{1}{2}(a_2 + 3a_3)$	$\frac{1}{2}(3a_2 + a_3)$	$-\frac{1}{2}(a_2 - a_3)$
$H^0(\wedge^2 T_Y)$ mixed	$\frac{1}{4}(a_0 + 9a_1 - 3a_2 - 3a_3)$	$\frac{1}{4}(a_0 - 3a_1 + a_2 - 3a_3)$	$\frac{1}{4}(a_0 - 3a_1 - 3a_2 + a_3)$	$\frac{1}{4}L_0(a)$

$H^{0,2}$ and the unmixed pieces of $\wedge^2 T_Y$: the block $K = (a_2, a_3)$ of P_a . At a rational exceptional class no entry vanishes, by the lemma on conjugates, except $b_{11} - b_{22}$, which leaves κ , and L_0 , which can be made nonzero by adding a multiple of π_0 .